STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: CNC-02 | TAG: 01191439

CNC-02: Haas VF-2SS CNC Vertical Machining Center

Facility Zone: Zone 2: CNC Machining (CNC) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

CRITICAL ENTANGLEMENT WARNING: STRICTLY NO GLOVES WHILE SPINDLE / BLADE IS ROTATING

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Glasses
2. Closed-Toe Shoes
3. NO GLOVES During Tool Loading or Spindle Operation

2. PRIMARY OPERATIONAL HAZARDS

1. 12,000 RPM high-speed spindle and 1,400 IPM high-velocity rapids
2. Unbalanced toolholder ejection at >=10,000 RPM
3. High-pressure coolant and sharp swarf

3. BEFORE OPERATION CHECKLIST

1. Run 20-minute Spindle Warm-Up program if machine has been idle >24 hours.
2. Do NOT run >=10,000 RPM (requires Grade G2.5 balanced toolholders not stocked in general crib).
3. Clean coolant tank intake filter & chip tray screen before starting cycle.

4. DURING OPERATION SAFETY RULES

1. Perform first-run verification at 5% rapid override with hand on E-Stop and Single Block + Distance-To-Go active.
2. Keep enclosure doors closed and interlock engaged.

5. AFTER USE & SHOWROOM CLEANUP

1. Wash down interior ONLY with E-Stop depressed.
2. Shake out cloth net primary coolant tray filter into chip bin.
3. NEVER use compressed air inside enclosure.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Depress E-Stop immediately; report any crash, alarm code, or tool breakage to Lab Supervisor before resetting.